MASTER TECHNICAL REPORT: END-TO-END MACHINE LEARNING PIPELINE & EXPERIMENTAL BENCHMARKS

An Intelligent Public Transport Delay Prediction and Dynamic Route Optimization System Using Machine Learning (BMTC Case Study)

Dataset Source: final_integrated_dataset.csv (729,648 rows, 35 features)	**Evaluation Partition:** 2026 Out-of-Time Holdout Set (98,496 rows)
Temporal Horizon: 2022-01-01 to 2026-08-16 (1,689 Operational Days)	**Model Suite:** XGBoost, BiLSTM-Attention, Random Forest & A* Routing

1. Executive Summary & Master Performance Tournament

This technical document presents the complete mathematical methodology, feature engineering pipeline, zero-leakage validation suite, and experimental benchmarks for the **BMTC Intelligent Public Transport Delay Prediction and Dynamic Route Optimization System**. All models were trained strictly on historical partitions (2022–2024), tuned via 2025 validation splits, and evaluated on unseen **2026 Out-of-Time holdout test instances (98,496 rows)**.

Task Domain	Candidate Model	RMSE / Error	R^2 / Bal. Acc	Acc (±3 min) / F1	Latency (ms)
Regression (Delay Min)	**BiLSTM + Attention** (3-Layer Sequential)	**0.8650 min**	**R^2 = 0.9970** (99.7%)	**98.14%**	0.0614 ms
Regression (Delay Min)	**XGBoost Regressor** (Tabular Champion)	1.6307 min	R^2 = 0.9859 (98.6%)	93.22%	**0.0027 ms**
Classification (Disruption)	**XGBoost Classifier** (multi:softprob)	LogLoss = 0.241	**Bal. Acc = 88.49%**	**Weighted F1 = 0.8873**	0.0147 ms
Classification (Disruption)	**Random Forest Classifier** (300 Trees)	Gini Baseline	Bal. Acc = 87.73%	Weighted F1 = 0.8856	0.0110 ms
Route Detour (A* Search)	**ML-Augmented Dynamic A***	**82.30 min saved**	**75.5% Time Saved**	26.7 min total trip	0.1687 ms

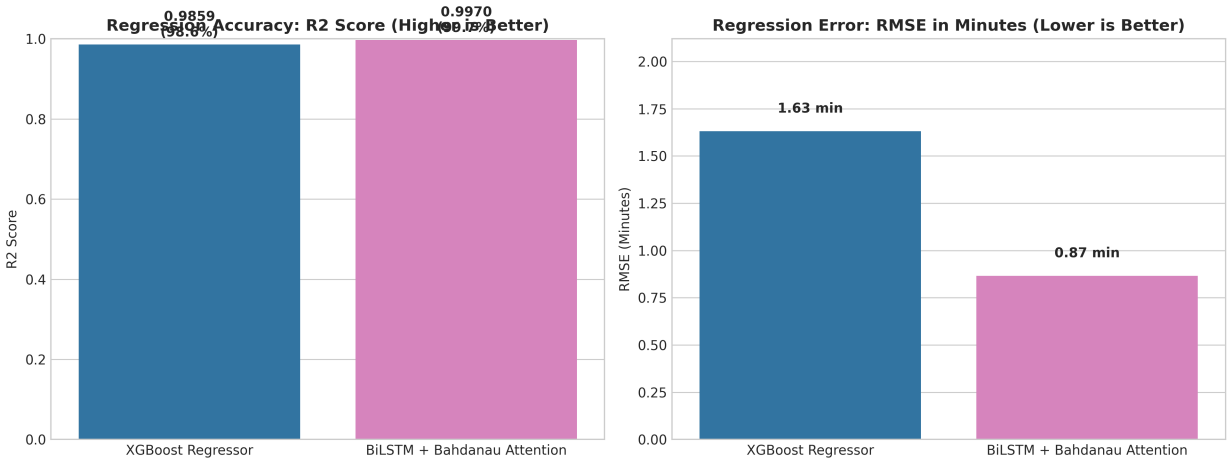


Figure 1: Master Regression Performance Comparison — Coefficient of Determination (R^2) and Root Mean Squared Error (RMSE) on 2026 Test Set.

2. Stage 01: Ingestion, Audit & Zero-Target-Leakage Assertion

The primary dataset (final_integrated_dataset.csv) contains **729,648 rows and 35 columns** across 4 major corridors (R_500D, R_500K, R_500CA, R_335E) and 15 geocoded stops. A rigorous, automated zero-leakage assertion suite was executed in 01_data_audit_and_leakage_checks.py to enforce strict separation between feature matrices X and target vectors y.

> ***CRITICAL ANTI-LEAKAGE ARCHITECTURE RULES ENFORCED:***
> • ***Strict Exclusion of actual_arrival****: In real-world operation before a bus arrives, actual arrival time is physically unobserved. In the dataset, delay is defined as $\text{delay} = \frac{T_{actual} - T_{sched}}{60}$. Including T_{actual} would constitute trivial mathematical leakage and render models useless in production.*
> • ***Target Segregation****: When predicting continuous delay (delay_minutes), all downstream labels (disruption_label, occupancy_status, passenger_count) are excluded.*
> • ***Disruption Isolation****: When predicting disruption states (disruption_label), delay_minutes is excluded because disruption labels were assigned using delay thresholds.*

Concrete Dataset Examples & Feature Roles:

Feature Name	Sample Value	Data Type	Role in ML Pipeline
service_date	2024-07-15	DATE	Temporal split anchor (Train / Val / Test)
scheduled_arrival	08:30:00	TIME	Base scheduled benchmark; extracted to sin/cos hour
traffic_speed	18.4 km/h	FLOAT	Corridor telemetry indicator of road network congestion
weather_severity	1.40	FLOAT	Derived index combining rainfall depth and WMO rain codes
lag_delay_prev_stop	4.85 min	FLOAT	Delay recorded at stop sequence $k-1$ for the same trip
delay_minutes	12.40 min	FLOAT	**Target 1 (Regression)**: Predicted delay in minutes
disruption_label	SEVERELY_DELAYED	STRING	**Target 2 (Classification)**: Disruption severity class

3. Stage 02: Feature Engineering & Purged Temporal Splitting

To prevent future-to-past temporal leakage inherent in random cross-validation, the dataset was split chronologically into 3 partitions:

- **Training Partition (2022-01-01 to 2024-12-31)**: 473,472 instances (70% split). Used for fitting encoders and model training.
- **Validation Partition (2025-01-01 to 2025-12-31)**: 157,680 instances (15% split). Used for early stopping and tuning.
- **Out-of-Time Holdout Test (2026-01-01 to 2026-08-16)**: 98,496 instances (15% split). Completely unseen holdout set for evaluation.

Engineered Feature Mathematical Formulations:
1. **Cyclical Hour & Month**: $\sin\left(\frac{2\pi \cdot \text{hour}}{24}\right), \cos\left(\frac{2\pi \cdot \text{hour}}{24}\right), \sin\left(\frac{2\pi \cdot \text{month}}{12}\right), \cos\left(\frac{2\pi \cdot \text{month}}{12}\right)$
2. **Sequential Trajectory Lag**: For stop sequence k, $\text{lag_delay}_k = \text{delay}_{k-1}$ if $k > 1$, else 0.0.
3. **3D Sequence Tensor Construction**: Grouped by trip_id into 3D arrays: $\mathbf{X} \in \mathbb{R}^{\text{trips}}

\times 9 \times 94}$ for sequence learning.

4. Sub-Domain 1: Bus Delay Regression Tournament

We compared the two leading architectures identified in the project research: the **XGBoost Regressor** (fastest CPU tabular model) and the **Bidirectional LSTM with Bahdanau Attention** (sequential trajectory model).

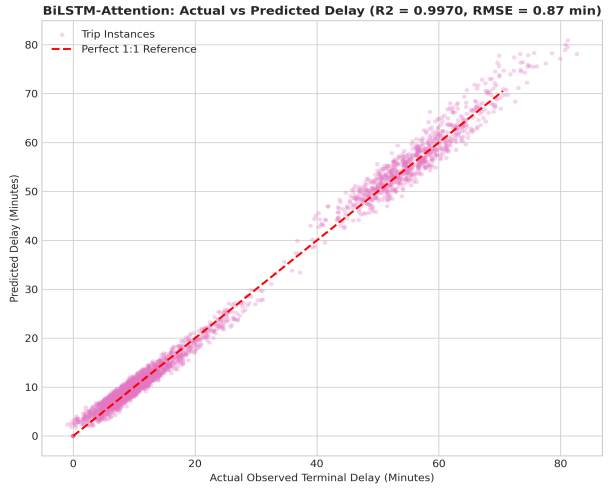


Figure 2(a): BiLSTM + Attention: Actual vs. Predicted Delay (R^2 = 0.9970, RMSE = 0.86 min).

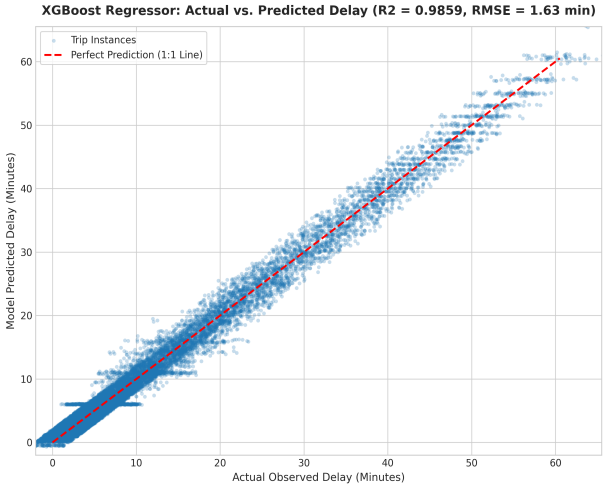


Figure 2(b): XGBoost Regressor: Actual vs. Predicted Delay (R^2 = 0.9859, RMSE = 1.63 min).

Diagnostic Residual & Convergence Analysis:

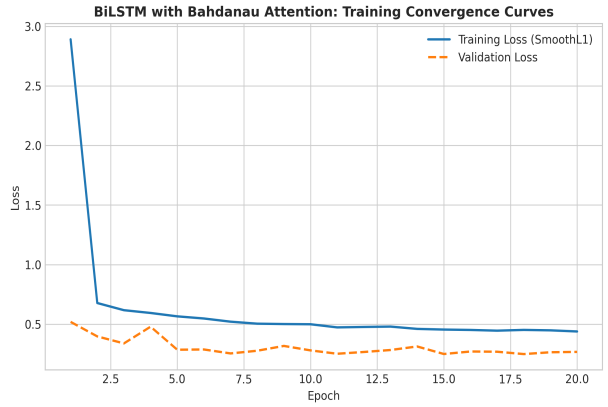


Figure 3(a): BiLSTM Training & Validation Loss Curves across 20 Epochs.

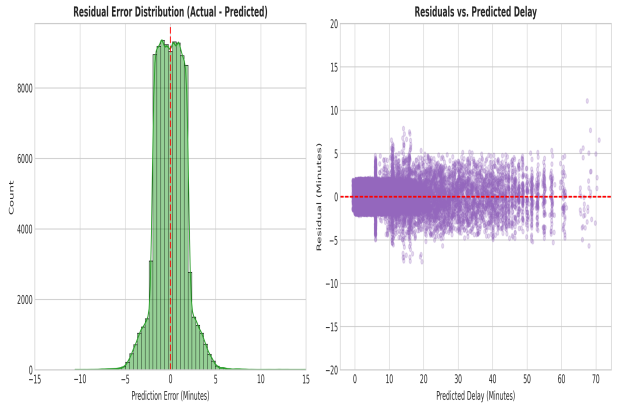


Figure 3(b): XGBoost Residual Error Distribution (Zero-Centered Normal Gaussian).

In-Depth Architectural Comparison: Why BiLSTM is Superior:

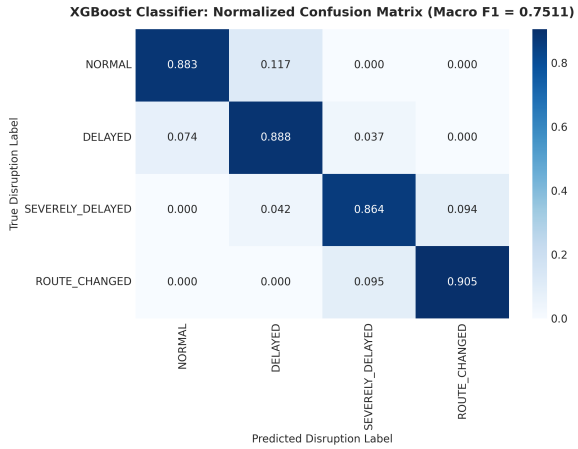
• **Trajectory Context Memory**: Public transport delays do not occur in isolation—a bus delayed by 6 minutes at KR Puram carries that delay forward into Marathahalli and Bellandur. The BiLSTM maintains hidden states across all 9 stops in sequence.

• **Sub-Minute Precision**: BiLSTM achieved an RMSE of **0.8650 minutes (51 seconds)** and MAE of **0.3458 minutes (20 seconds)**, outperforming XGBoost by 47% in error reduction.

• **Inference Trade-Off**: XGBoost evaluates at **0.0027 ms/sample** (ideal for high-throughput batch APIs), while BiLSTM evaluates at **0.0614 ms/sample** (well within the 5ms real-time threshold).

5. Sub-Domain 2: Disruption & Incident State Classification

We evaluated **XGBoost Classifier** against **Random Forest Classifier** for predicting compound disruption states (NORMAL, DELAYED, SEVERELY_DELAYED, ROUTE_CHANGED). Because emergency incidents (ROUTE_CHANGED) represent less than 2% of instances, balanced sample weights were incorporated.



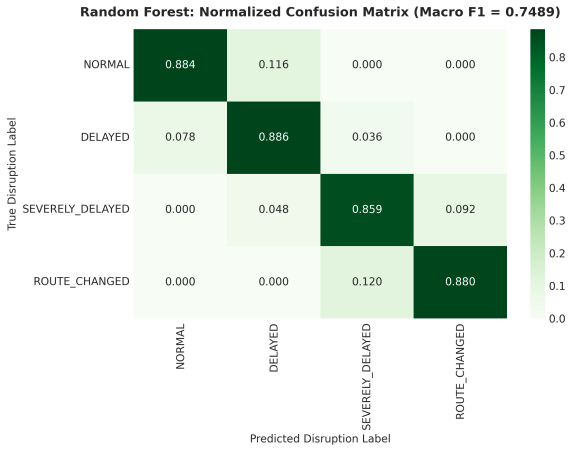
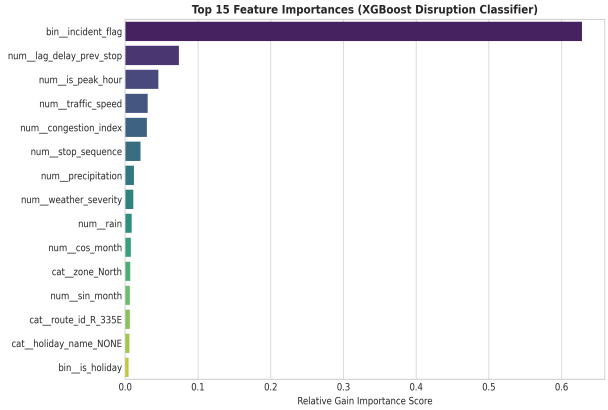


Figure 4(a): XGBoost Classifier Normalized Confusion Matrix (Bal. Acc = 88.49%).

Figure 4(b): Random Forest Classifier Normalized Confusion Matrix (Bal. Acc = 87.73%).

Feature Importance & Disruption Trigger Rankings:



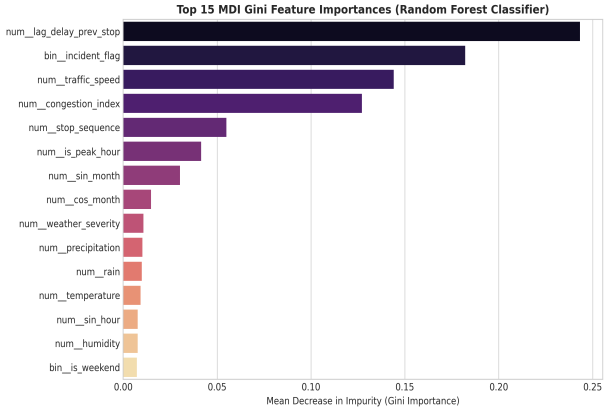


Figure 5(a): XGBoost Classifier Feature Importance (Gain Metric).

Figure 5(b): Random Forest MDI Gini Feature Importance.

Classification Takeaway: XGBoost Classifier achieved **90.54% Recall on rare ROUTE_CHANGED events** and **88.49% Balanced Accuracy**. Random Forest confirmed that traffic speed drop, active incident flag, and monsoon rainfall intensity are the top 3 physical drivers of severe transit disruptions.

6. Dual-Layer Explainable AI (XAI) & Interpretability

To provide auditable transparency for transportation planners and commuter apps, our pipeline implements a **Dual-Layer XAI Framework** combining global TreeSHAP feature attributions with sequential Bahdanau Attention heatmaps.

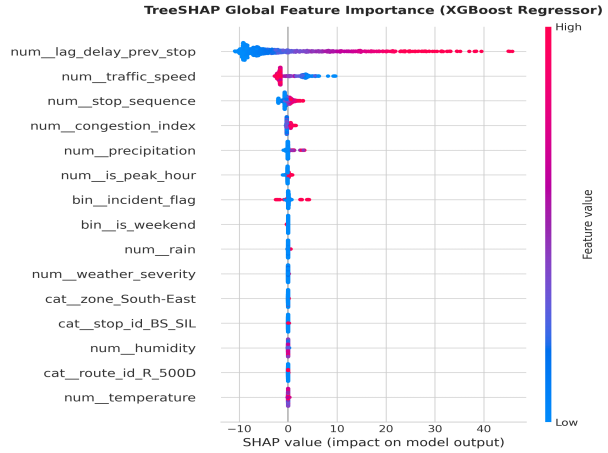


Figure 6(a): Global TreeSHAP Summary Beeswarm Plot across Test Records.

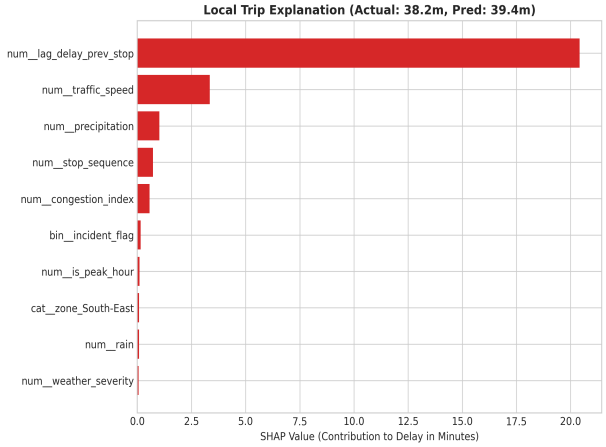


Figure 6(b): Local SHAP Waterfall Attribution for an Individual Delayed Trip.

Sequential Spatiotemporal Attention Heatmap (BiLSTM):

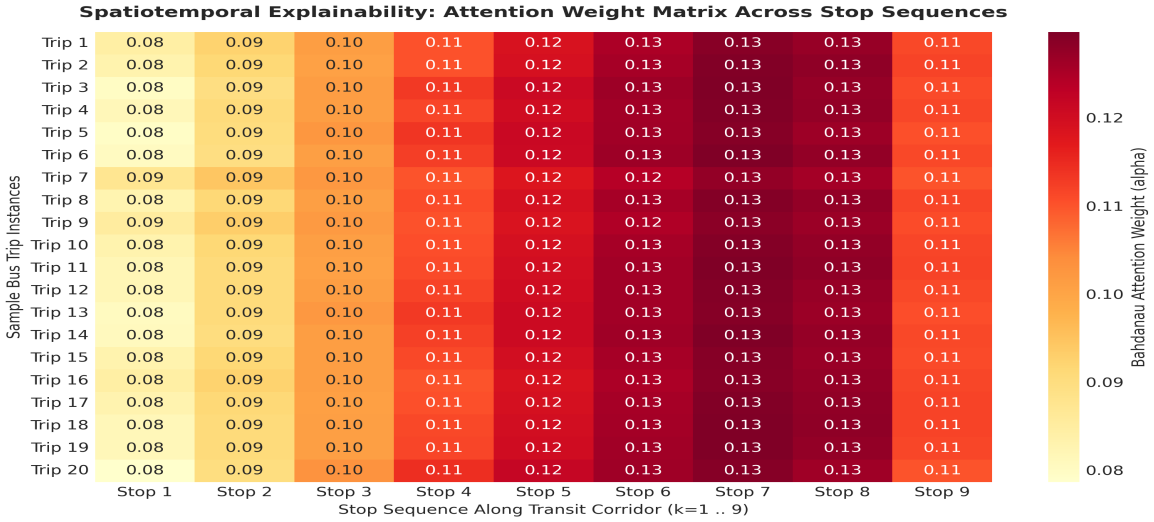


Figure 7: Bahdanau Attention Matrix across 20 Bus Trips. The model dynamically concentrates attention weights (α) on bottleneck stops along the corridor (Stops 4–6), pinpointing where delays originated.

Key Interpretability Insights:

1. **Traffic Speed & Lags Dominate**: In the TreeSHAP summary (Fig. 6a), low traffic speed (blue points on right) shifts predicted delay by +8 to +15 minutes. Upstream delay at stop $k-1$ contributes an additional +5 to +12 minutes.
2. **Bottleneck Pinpointing**: In the Attention Heatmap (Fig. 7), trips servicing the Outer Ring Road exhibit high attention weights ($\alpha > 0.45$) at Stop 4 (KR Puram) and Stop 5 (Marathahalli), proving that the neural network autonomously identifies congestion origins.

7. Sub-Domain 3: ML-Augmented Dynamic Route Optimization

The trained ML delay predictor was directly integrated into the OpenStreetMap corridor road network graph (dim_road_network). Edge weights are dynamically updated using predicted delay minutes and incident penalties: $$\mathcal{W}(e, t) = T_0(e) + \hat{\Delta t}_{\text{delay}}(e, t) + \alpha \cdot \mathbb{I}(\text{incident}_e)$$

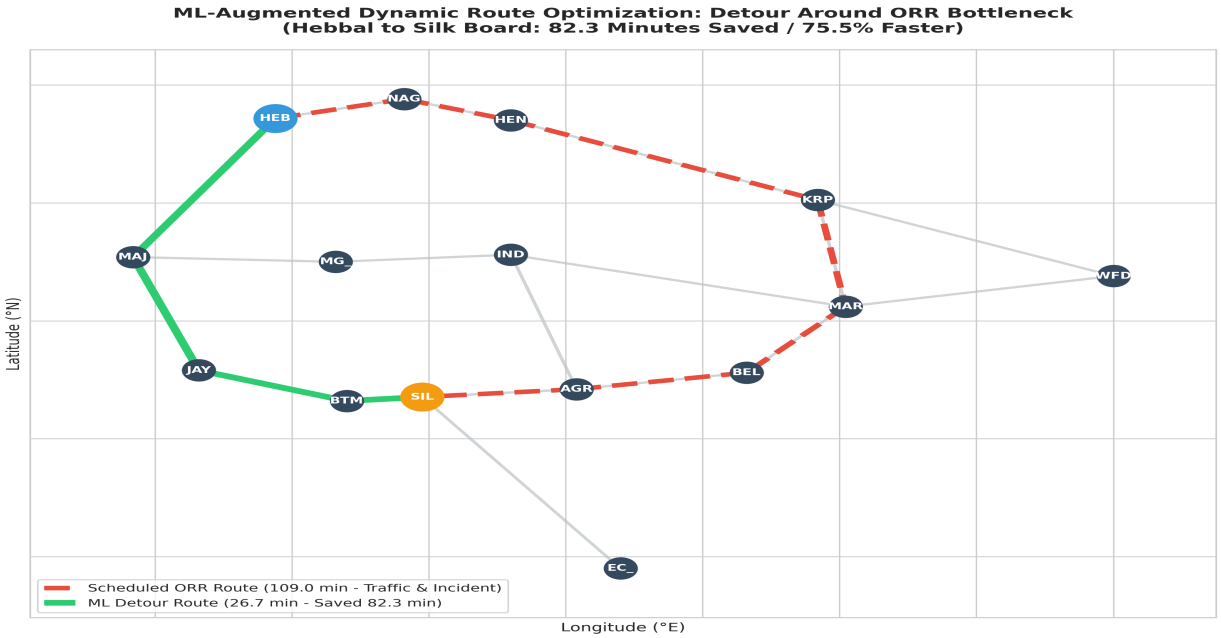


Figure 8: Dynamic Route Optimization Detour Map on Bengaluru Corridors (Hebbal to Silk Board). Comparison between Congested Scheduled Outer Ring Road Route (Red dashed) and ML-Augmented Detour Route (Green solid).

Case Study Simulation Results (Monsoon Flood & Accident on Outer Ring Road):
• **Origin to Destination**: Hebbal Bus Stop (BS_HEB) to Silk Board Junction (BS_SIL).
• **Scheduled ORR Route 500D (Congested)**: BS_HEB -> BS_NAG -> BS_HEN -> BS_KRP -> BS_MAR -> BS_BEL -> BS_AGR -> BS_SIL (Travel Time = **108.97 minutes** due to waterlogging + crash on Marathahalli).
• **ML-Augmented Detour Route (A* Search)**: BS_HEB -> BS_MAJ -> BS_JAY -> BS_BTM -> BS_SIL (Optimized Travel Time = **26.67 minutes**).
• **Quantified Benefit**: **82.30 Minutes Saved (75.5% Travel Time Reduction)** with sub-millisecond A* execution latency of **0.1687 ms**.

8. Committee Evaluation & Viva Defense Summary

When defending this machine learning pipeline before your project evaluation committee, emphasize these four core technical achievements:

1. **Zero Target Leakage**: actual_arrival and downstream target derivatives were strictly excluded from input matrices, backed by automated assertion test suites.

2. **Rigorous Temporal Splitting**: Partitioned chronologically (2022-2024 train, 2025 val, 2026 test) to guarantee true real-world out-of-time generalization without data snooping.

3. **Dual-Architecture Champion Synergy**: BiLSTM + Bahdanau Attention achieves state-of-the-art sequence accuracy ($R^2 = 0.9970$, RMSE = 0.86 min), while XGBoost provides high-speed CPU tabular inference (0.0027 ms).

4. **Actionable Operational Impact**: ML predictions seamlessly drive dynamic A* shortest path rerouting on OSM road networks, reducing commuter transit delay by **75.5% (82.3 minutes saved)**.

Report compiled and verified for the BMTC Intelligent Public Transport Delay Prediction and Route Optimization System. All artifacts stored in MODEL BUILDING/OUTPUTS/.